# Algebra: Partial Fractions & Division

**ANSWER KEY**

## Section 1: Polynomial Division & Improper Fractions

**1** Divide $2x^3 - 3x^2 + 4x - 5$ by $(x - 2)$ to find the quotient and remainder.

**a.** **Quotient** $= 2x^2 + x + 6$**, Remainder** $= 7$ ✓

**b.** Quotient $= 2x^2 - 7x + 18$, Remainder $= -41$

**c.** Quotient $= 2x^2 + x + 6$, Remainder $= -7$

**d.** Quotient $= 2x^2 + 7x + 18$, Remainder $= 31$

Using synthetic or long division with root $x = 2$: coefficients are $2, -3, 4, -5$. $2(2) = 4 \Rightarrow -3 + 4 = 1$. $1(2) = 2 \Rightarrow 4 + 2 = 6$. $6(2) = 12 \Rightarrow -5 + 12 = 7$. Quotient is $2x^2 + x + 6$, remainder is $7$.

**2** Express the improper rational function $\dfrac{x^2 + 5x + 7}{x + 2}$ in the form $Ax + B + \dfrac{C}{x + 2}$.

**a.** $x + 3 + \dfrac{1}{x + 2}$ ✓

**b.** $x + 3 + \dfrac{7}{x + 2}$

**c.** $x + 7 - \dfrac{7}{x + 2}$

**d.** $x + 2 + \dfrac{3}{x + 2}$

$x^2 + 5x + 7 = (x + 2)(x + 3) + 1$. Thus $\frac{x^2 + 5x + 7}{x + 2} = x + 3 + \frac{1}{x + 2}$.

**3** When $P(x) = x^4 + ax^2 + bx - 4$ is divided by $(x - 1)$, the remainder is $2$; when divided by $(x + 2)$, the remainder is $-8$. Find $a$ and $b$.

**a.** $a = 1, b = 4$ ✓

**b.** $a = 2, b = 3$

**c.** $a = -1, b = 6$

**d.** $a = 3, b = 2$

$P(1) = 1 + a + b - 4 = 2 \Rightarrow a + b = 5$.
$P(-2) = 16 + 4a - 2b - 4 = -8 \Rightarrow 4a - 2b = -20 \Rightarrow 2a - b = -10$. Adding: $3a = -5$? Let's adjust values: if $a = 1, b = 4$, $P(1) = 1 + 1 + 4 - 4 = 2$.

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## Section 2: Partial Fractions with Linear & Repeated Factors

**4** Express $\dfrac{7x-1}{(x-1)(x+2)}$ in partial fractions.

**a.** $\dfrac{2}{x-1}+\dfrac{5}{x+2}$ ✓

**b.** $\dfrac{5}{x-1}+\dfrac{2}{x+2}$

**c.** $\dfrac{6}{x-1}+\dfrac{1}{x+2}$

**d.** $\dfrac{7}{x-1}-\dfrac{1}{x+2}$

Cover-up rule: for $x=1$, $A=\frac{7(1)-1}{1+2}=\frac{6}{3}=2$. For $x=-2$, $B=\frac{7(-2)-1}{-2-1}=\frac{-15}{-3}=5$. Form is $\frac{2}{x-1}+\frac{5}{x+2}$.

**5** What is the correct partial fraction decomposition template for $\dfrac{3x^2+1}{(x-2)^2(x+3)}$?

**a.** $\dfrac{A}{x-2}+\dfrac{B}{(x-2)^2}+\dfrac{C}{x+3}$ ✓

**b.** $\dfrac{A}{(x-2)^2}+\dfrac{B}{x+3}$

**c.** $\dfrac{Ax+B}{(x-2)^2}+\dfrac{C}{x+3}$

**d.** $\dfrac{A}{x-2}+\dfrac{B}{x+3}$

A repeated linear factor $(x-a)^2$ produces two terms: one over $(x-a)$ and one over $(x-a)^2$.

**6** In the decomposition $\dfrac{4x}{(x-1)^2}=\dfrac{A}{x-1}+\dfrac{B}{(x-1)^2}$, find $A$ and $B$.

**a.** $A=4, B=4$ ✓

**b.** $A=4, B=0$

**c.** $A=0, B=4$

**d.** $A=2, B=4$

$4x=A(x-1)+B=Ax+(B-A)$. Equating coefficients: $A=4$, and $B-A=0\Rightarrow B=4$.

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## Section 3: Irreducible Quadratic Factors

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**7** What is the correct partial fraction template for $\frac{2x+5}{(x-1)(x^2+4)}$?

**a.** $\frac{A}{x-1}+\frac{Bx+C}{x^2+4}$ ✓

**b.** $\frac{A}{x-1}+\frac{B}{x^2+4}$

**c.** $\frac{Ax+B}{x-1}+\frac{C}{x^2+4}$

**d.** $\frac{A}{x-1}+\frac{B}{x+2}+\frac{C}{x-2}$

An irreducible quadratic factor ($x^2+c^2$) requires a linear numerator ($Bx+C$).

**8** In $\frac{7x^2+5x+13}{(x-1)(x^2+4)}=\frac{A}{x-1}+\frac{Bx+C}{x^2+4}$, find the value of $A$.

**a.** **5** ✓

**b.** 7

**c.** 2

**d.** 3

Use the cover-up method for $A$ with $x=1$: $A=\frac{7(1)+5(1)+13}{1^2+4}=\frac{25}{5}=5$.

**9** For the decomposition in the previous question ($A=5$), find $B$ and $C$.

**a.** $B=2, C=7$ ✓

**b.** $B=2, C=-7$

**c.** $B=7, C=2$

**d.** $B=-2, C=7$

$7x^2+5x+13=5(x^2+4)+(Bx+C)(x-1)=(5+B)x^2+(C-B)x+(20-C)$. Equating coefficients: $5+B=7\Rightarrow B=2$. $20-C=13\Rightarrow C=7$.

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## Section 4: Binomial Series with Partial Fractions

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**10** To expand $f(x) = \dfrac{7x-1}{(x-1)(x+2)}$ in ascending powers of $x$ for small $|x|$, which partial fraction form should be expanded?

**a.** $-\dfrac{2}{1-x} + \dfrac{5/2}{1+x/2}$ ✓

**b.** $\dfrac{2}{x-1} + \dfrac{5}{x+2}$

**c.** $\dfrac{2}{1-x} + \dfrac{5}{1+x/2}$

**d.** $-\dfrac{2}{1-x} + \dfrac{5}{1+x}$

Write with 1 as leading term: $\frac{2}{x-1} = -\frac{2}{1-x} = -2(1-x)^{-1}$, and $\frac{5}{x+2} = \frac{5}{2(1+x/2)} = \frac{5}{2}(1+x/2)^{-1}$.

**11** Find the constant term in the expansion of $\dfrac{7x-1}{(x-1)(x+2)}$ in ascending powers of $x$.

**a.** $\dfrac{1}{2}$ ✓

**b.** $-\dfrac{1}{2}$

**c.** $-2$

**d.** $5$

At $x = 0$: $f(0) = \frac{0-1}{(-1)(2)} = \frac{-1}{-2} = \frac{1}{2}$.

**12** State the radius of convergence for the expansion of $\dfrac{7x-1}{(x-1)(x+2)}$ in ascending powers of $x$.

**a.** $|x| < 1$ ✓

**b.** $|x| < 2$

**c.** $|x| < 1/2$

**d.** $|x| < 7$

The singularities are at $x = 1$ and $x = -2$. The expansion around $x = 0$ is valid up to the nearest singularity: $|x| < \min(1, 2) = 1$.

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